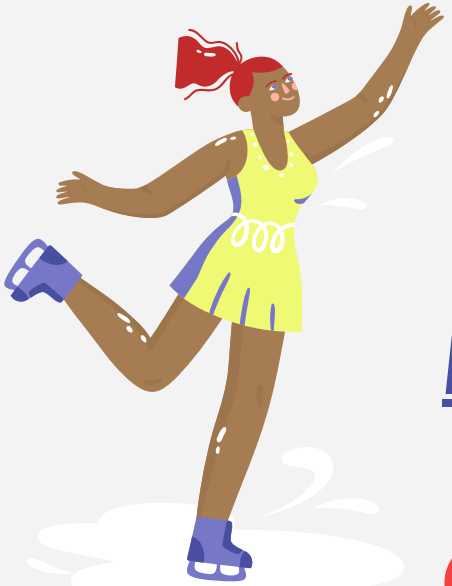


Winter Olympics

Medal Data Analysis

By: Isabelle Adams & Naomi Tefera

Research Questions

01

- Is there a correlation between being from the host country and whether or not an athlete wins a medal? Does home advantage actually show up in the data?

02

- Are there any countries that are dominant in a specific sport? And are these the same countries that dominate overall, or do smaller nations punch above their weight in certain events?

03

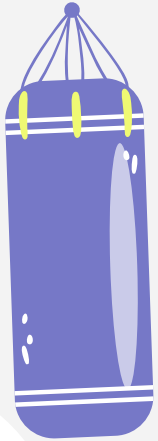
- Has the global distribution of medals changed over time? Are traditional powerhouses still dominating, or are new countries emerging as competitors?

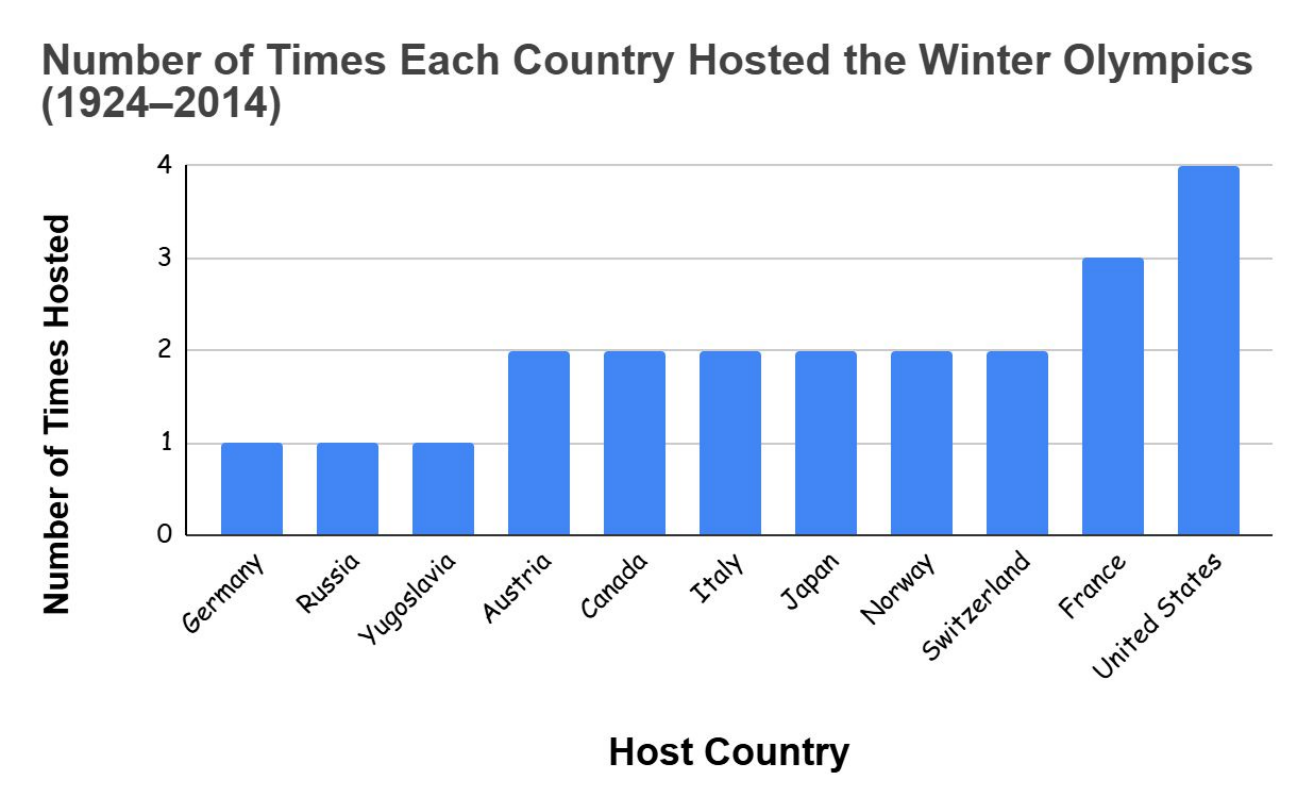
Dataset Overview

- Winter Olympics data from 1924-2014
- Includes medal counts, countries, sports, and host locations
- Cleaned & standardized country codes
- Pivot tables & visualizations to explore trends

01

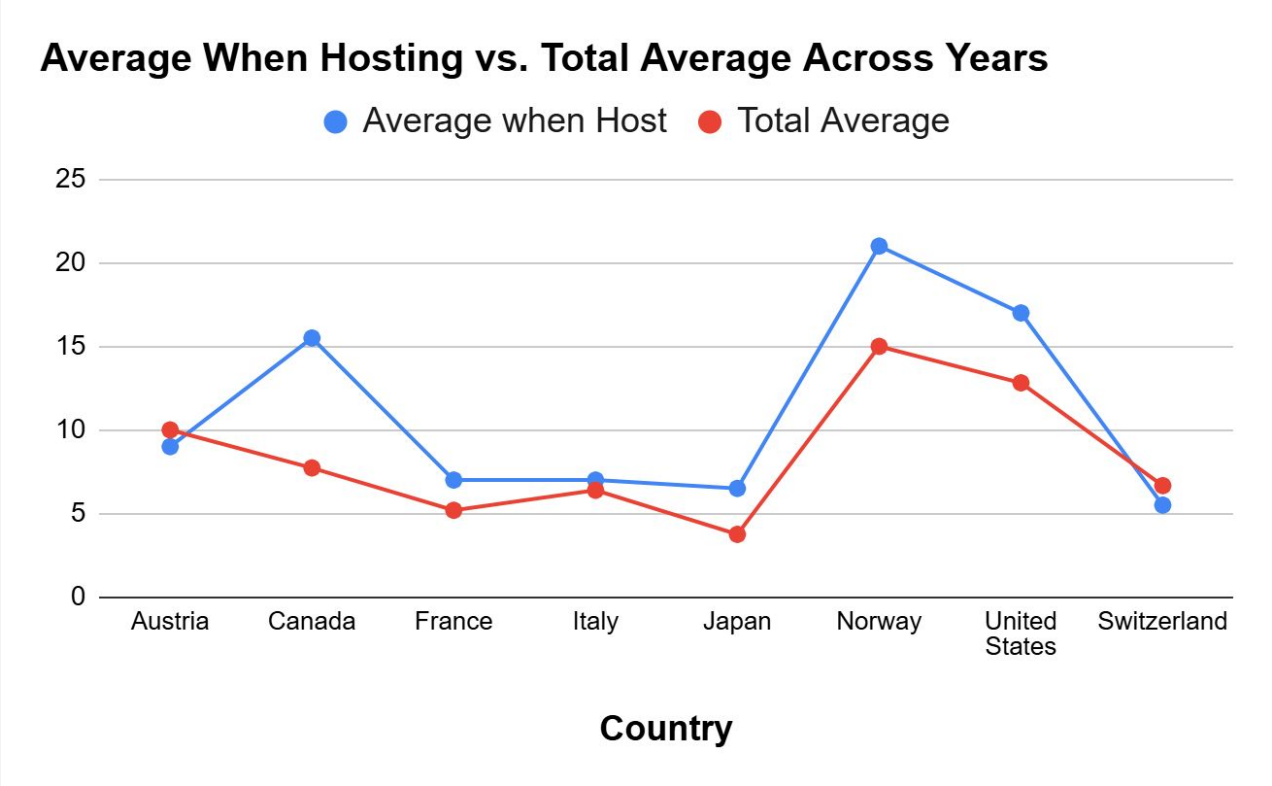
Is There A Hometown Advantage For Athletes?




Number of Times Each Country Hosted the Winter Olympics (1924–2014)
Number of Times Hosted
0
1
2
3
4
Germany
Russia
Yugoslavia
Austria
Canada
Italy
Japan
Norway
Switzerland
France
United States
Host Country

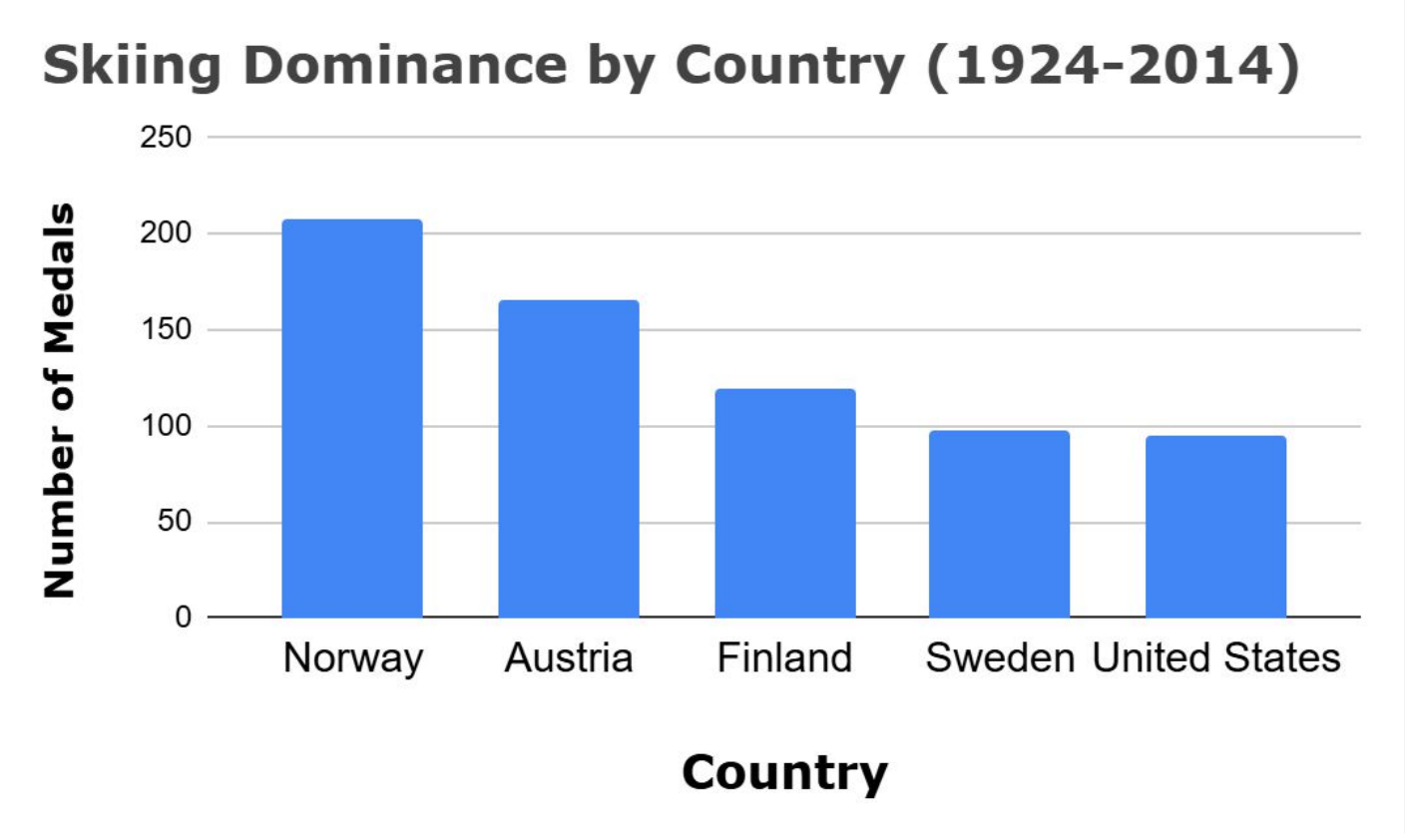
Testing The Host Advantage

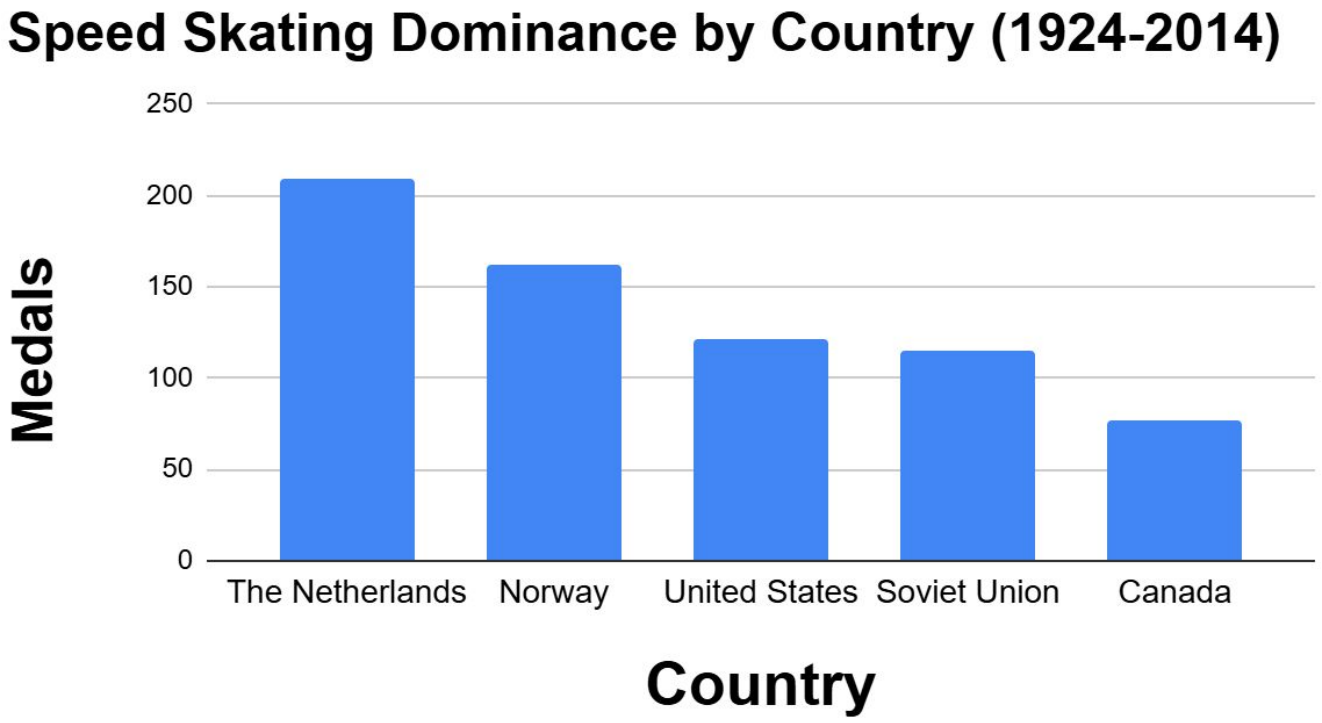
Hypothesis



02

Which Countries Dominate Specific Winter Olympic Sports?

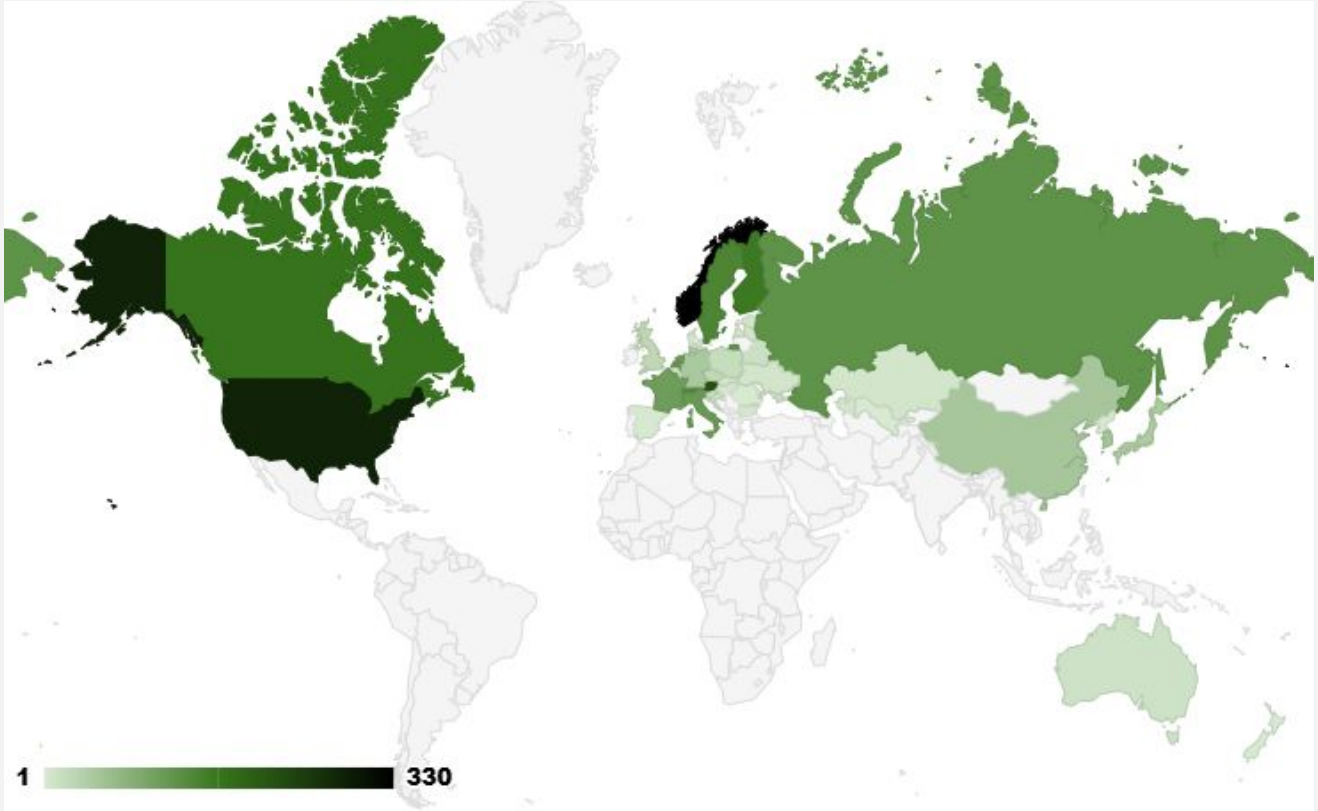

Skiing Dominance by Country (1924-2014)
Number of Medals
250
200
150
100
50
0
Norway
Austria
Finland
Sweden
United States
Country


Speed Skating Dominance by Country (1924-2014)
Medals
250
200
150
100
50
0
The Netherlands
Norway
United States
Soviet Union
Canada
Country

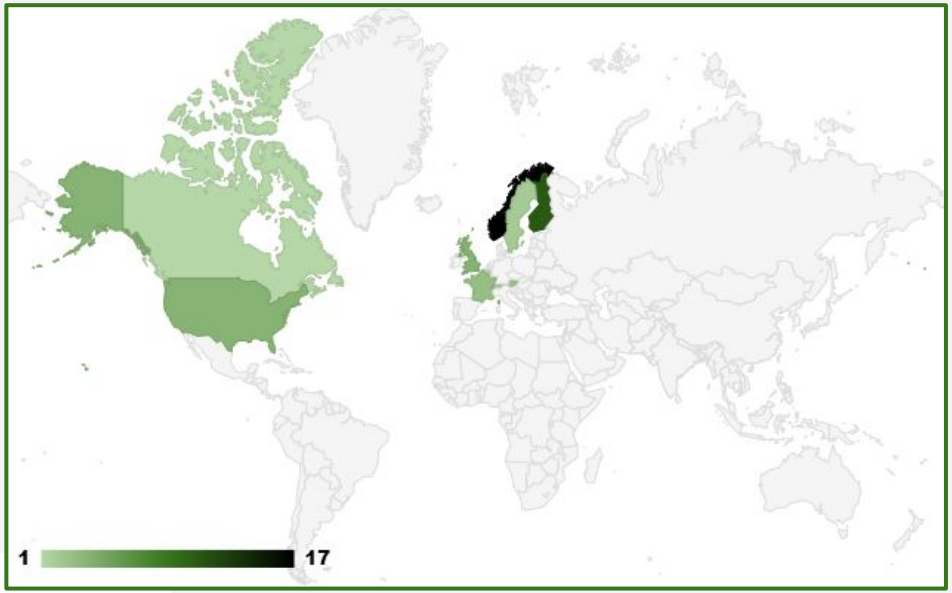
03

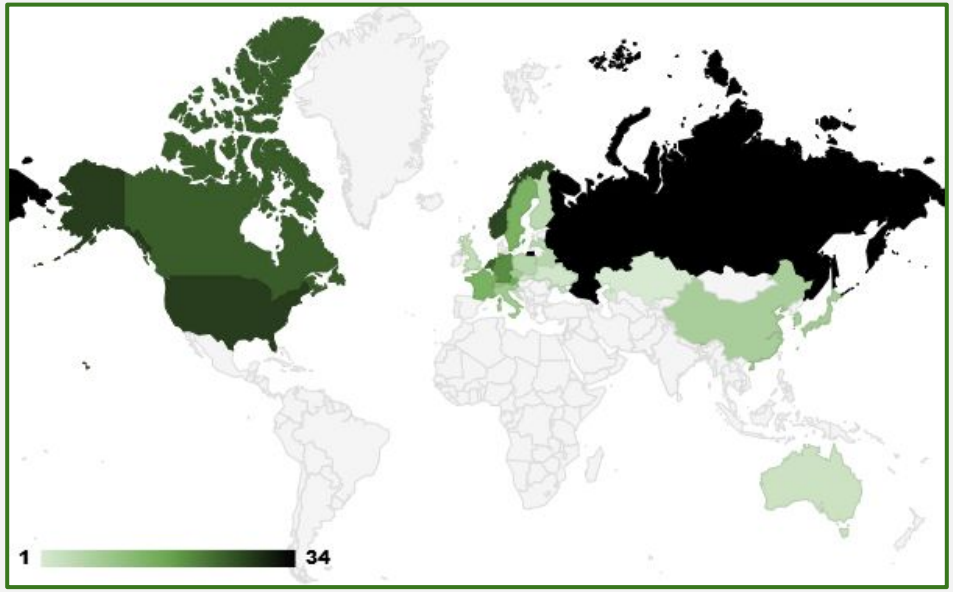
Has The Global Distribution Changed Over Time?

Total Medal Distribution



Medal Comparison (1924 vs. 2014)





Number of Countries Winning Medals Over Time
Number of Countries
50
40
30
20
10
0
1924
1934
1944
1954
1964
1974
1984
1994
2004
2014
Year

Overall Conclusion

Hosting Patterns

- Small number of repeated host countries
- Host countries win more medals

Sport-Specific Dominance

- Speed Skating has largest medal pool (1050)
- The Netherlands dominates speed skating, despite being smaller overall medal country

Global Medal Distribution

- In 1924, handful of countries won medals
- By 2014, medal-winning countries expanded

Sources

- https://www.kaggle.com/datasets/hassanjameelahmed/olympic-dataset
- https://docs.google.com/spreadsheets/d/1ngUopctktKmq41OBZpk0buCO3qrMhsHpxKkYpKzhZcI/edit?usp=sharing